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Efficient kNN Classification With Different Numbers of Nearest Neighbors

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***Abstract*—*k* nearest neighbor (kNN) method is a popular classification method in data mining and statistics because of its simple implementation and significant classification performance. However, it is impractical for traditional kNN methods to assign a fixed *k* value (even though set by experts) to all test samples. Previous solutions assign different *k* values to different test samples by the cross validation method but are usually time-consuming. This paper proposes a kTree method to learn different optimal *k* values for different test/new samples, by involving a training stage in the kNN classification. Specifically, in the training stage, kTree method first learns optimal *k* values for all training samples by a new sparse reconstruction model, and then constructs a decision tree (namely, kTree) using training samples and the learned optimal *k* values. In the test stage, the kTree fast outputs the optimal *k* value for each test sample, and then, the kNN classification can be conducted using the learned optimal *k* value and all training samples. As a result, the proposed kTree method has a similar running cost but higher classification accuracy, compared with traditional kNN methods, which assign a fixed *k* value to all test samples. Moreover, the proposed kTree method needs less running cost but achieves similar classification accuracy, compared with the newly kNN methods, which assign different *k* values to different test samples. This paper further proposes an improvement version of kTree method (namely, k*Tree method) to speed its test stage by extra storing the information of the training samples in the leaf nodes of kTree, such as the training samples located in the leaf nodes, their kNNs, and the nearest neighbor of these kNNs. We call the resulting decision tree as k*Tree, which enables to conduct kNN classification using a subset of the training samples in the leaf nodes rather than all training samples used in the newly kNN methods. This actually reduces running cost of test stage. Finally, the experimental results on 20 real data sets showed that our proposed methods (i.e., kTree and k*Tree) are much more efficient than the compared methods in terms of classification tasks.**

***Index Terms*—Decision tree, *k* nearest neighbor (kNN) classification, sparse coding.**

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I. Introduction

DIFFERENT from model-based methods which first learn models from training samples and then predict test samples with the learned model [1]–[6], the model-free *k* nearest neighbors (kNNs) method does not have training stage and conducts classification tasks by first calculating the distance between the test sample and all training samples to obtain its nearest neighbors and then conducting kNN classification[1] (which assigns the test samples with labels by the majority rule on the labels of selected nearest neighbors). Because of its simple implementation and significant classification performance, kNN method is a very popular method in data mining and statistics and thus was voted as one of top ten data mining algorithms [7]–[13].

Previous kNN methods include: 1) assigning an optimal *k* value with a fixed expert-predefined value for all test samples [14]–[19] and 2) assigning different optimal *k* values for different test samples [18], [20], [21]. For example, Lall and Sharma [19] indicated that the fixed optimal-*k*-value for all test samples should be $k = \sqrt{n}$ (where $n > 100$ and n is the number of training samples), while Zhu *et al.* [21] proposed to select different optimal *k* values for test samples via tenfold cross validation method. However, the traditional kNN method, which assigns fixed kNNs to all test samples (fixed kNN methods for short), has been shown to be impractical in real applications. As a consequence, setting an optimal-*k*-value for each test sample to conduct kNN classification (varied kNN methods for short) has been becoming a very interesting research topic in data mining and machine learning [22]–[29].

A lot of efforts have been focused on the varied kNN methods, which efficiently set different optimal-*k*-values to different samples [20], [30], [31]. For example, Li *et al.* [32] proposed to use different numbers of nearest neighbors

[1]In this paper, we call all kinds of methods, which use the labels of kNNs to classify the test samples, as kNN methods, such as fixed kNN methods and varied kNN methods. The kNN methods usually includes two steps, i.e., the setting of *k* value and the kNN classification. The *k* value can be set by experts in the fixed kNN methods while is learned (optimal *k* value for short) in the varied kNN methods. The kNN classification represents to use the majority rule of kNNs of the test sample to assign it with labels.